Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/06/2015
PI Name:	Hermanson, James C Ph.D.		
Project Title:	Stability, Cellular Structure and Heat transfer of Evaporating Films in Normal and Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Space Biology Special Category:	None		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Jeffrey (Michigan Technological University)		
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Task Description:

This program studies fundamental film evaporation phenomena under normal- and reduced-gravity conditions. Experimental tests, combined with detailed stability analysis, are employed to achieve an improved understanding of the fluid physics responsible for film instability, changes in the film structure, and the corresponding implications for heat transfer, and how those are impacted by the removal of buoyancy in reduced gravity. The mass loss and vapor recoil due to evaporation and potential changes in surface temperature owing to phase change are expected to play a de-stabilizing role for films in reduced gravity, where surface-tension driven convection can become important. The impacts of these mechanisms on the stability and structure for both buoyant and non-buoyant films are examined by the systematic variation of saturation temperature by changing the system pressure and the wall temperature. Both steady-state and transient film evaporation phenomena are studied.

A comprehensive suite of instrumentation is used for the systematic examination of the fluid physics described above. These diagnostics include temperature and heat transfer measurements as well as ultrasonic measurement of film thickness. Optical techniques are employed to determine the features of the resulting instabilities, variations in surface temperature, and the evolution of the cellular structure within the evaporating films.

In parallel with the experimental program, a stability analysis is performed to model the film evaporation and the development of interfacial disturbances. The analysis will be used both to aid in the interpretation of the ground-based results, as well as to shed important insight into the evaporating film behavior to be expected under long-term, microgravity conditions. The proposed ground-based investigation will conclude with the recommendation of a preliminary microgravity flight experiment design to NASA.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The high heat flux associated with phase change is an important consideration in spacecraft thermal control applications. Devices such as capillary pumped loops and heat pipes depend critically on evaporation and the successful utilization of these devices in space applications require a detailed understanding of evaporation phenomena under reduced gravity conditions. Other important technologies related to space exploration requiring a more in-depth understanding of evaporation include humidity control of cabin environment, water recovery from waste sources, oxygen production, water management in power systems such as fuel cells, and in-situ resource utilization for fuel production. Understanding evaporation phenomena is critical to the development and optimization of these space technologies that enable human exploration of space.
In addition to space applications, evaporation is important to many terrestrial technologies from drying applications in the pulp and paper industry to coating deposition on fiber optics. Many of these technologies, while reliable, are not optimized for lack of a thorough understanding of evaporation phenomena. International competition is driving a need beyond consistent, reliable operation of manufacturing and processing technologies towards more optimized operations. Research on evaporation mechanisms may assist in this optimization.

Experimental studies of films evaporating under transient conditions suggest a non-trivial change in the rate of heat transfer with time. Comparison of such transient films with those of steady, evaporating films of similar thickness and superheat levels indicates that the heat transfer associated with the transient films can exceed those of steady-state evaporating films by a factor of two or more. This points to the possible practical utility of modulating the evaporation rate to increase the rate of heat rejection and transport of waste heat.

Stability analysis and numerical modeling suggest that under zero-gravity conditions, evaporating films can undergo total film destabilization and the formation of thermocapillary structures, a potentially important consideration in space applications. These instabilities are not predicted under normal-gravity conditions. Experimental results suggest that the equilibrium condition at the evaporating surface largely suppresses surface temperature variation, effectively eliminating thermocapillary-driven instability in normal gravity. The stability analysis indicates that this may not be the case for evaporating films in zero gravity.

The use of ultrasound for the dynamic monitoring of film thickness and growth rate developed as part of this effort has potentially widespread use for the determination of film behavior in a variety of devices that employ two-phase flow, such as power stations and closed-loop refrigeration cycles, evaporators, and condensers.

Task Progress:

This experimental and numerical study of film evaporation in normal- and reduced gravity examined the interfacial film stability, the evolution of the convective structures within the evaporating film, and the implications for heat transfer. Films typically a few millimeters in thickness evaporated into their own vapor on a planar, temperature-controlled surface in an enclosed test cell. The experiments were conducted under transient conditions that resulted from the impulsive application of superheating and the modulation of the superheat level within the film by varying the system pressure. The impulsive-superheating experiments were conducted both in the laboratory and in reduced gravity on board a parabolic-trajectory aircraft; the pressure-modulated experiments were conducted in the laboratory only. Diagnostics included temperature and pressure measurements as well as ultrasonic measurement of the film thickness, from which the rate of heat transfer at the evaporating interface was determined. Optical imaging was employed to determine changes in the convective structure within the film.
The normal-gravity experiments confirmed the presence of two distinct peaks in heat flux under transient conditions. The first peak is associated with a conduction-dominated phase of transient superheating; the second, by a marked changed in the developing convective structure. The characteristic times associated with these two events, in many cases, decreased with the progression of the pressure-modulated cycles. This behavior can be largely attributed to the developing thermal profile within the film under these nonstationary conditions. The convective structure appeared under these cyclical evaporation conditions exhibited a structure and range of wavelengths different from that observed during quasi-steady film evaporation. In some cases convective structure was observed to persist in the pressure-modulated film even during intervals in which the superheating was nulled and the evaporation was quenched. Although a short-term increase in the heat flux at the evaporating surface could be attained by using the pressure-modulated technique, over multiple cycles the time-averaged heat flux generally tended to approach that of quasi-steady films.

Film-evaporation experiments in reduced gravity were generally not successful due to significant intervals of upwards (“positive”) gravity during each zero-gravity parabola, which rendered the bulk of the film unstable. Other significant disturbances due to g-level variations on board the parabolic-trajectory aircraft and background vibrations generally precluded the acquisition of useable film-evaporation data under reduced-gravity conditions. This points to the need for

	a long-term microgravity platform (such as the International Space Station--ISS) for the study of the behavior of evaporating films under zero-g conditions. The experiments were supplemented by stability analysis of fluid layers undergoing evaporation; the results of that effort were presented in a previous report.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
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Articles in Peer-reviewed Journals	Narendranath AD, Hermanson JC, Kolkka RW, Struthers AA, Allen JS. "The effect of gravity on the stability of an evaporating liquid film." Microgravity Science and Technology. 2014 Oct;26(3):189-99. http://dx.doi.org/10.1007/s12217-014-9395-8 , Oct-2014
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